KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 11

YEAR 9 MATHEMATICS

Coordinate geometry and graphs

10 lessons · three pages each · Easy drill, then Medium and worded problems

11.1 The length of an interval
11.2 The midpoint of an interval
11.3 The gradient of an interval
11.4 Graphing linear equations
11.5 The gradient–intercept equation
11.6 Finding the equation of a line
11.7 Solving linear equations graphically
11.8 Linear modelling
11.9 Graphing parabolas
11.10 Parabolas in real life

NAME _____ CLASS _____ TEACHER _____

Artwork: silver seahorse tails deep in Seahorse Valley, coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **three pages**. Read the **learning intention**, copy your teacher's notes into the **Notes** box, and tick each **success criterion** when you can do it.

2. Pages 1 and 2 are **Easy**: small steps, one round at a time. Page 3 is **Medium**, then three **worded problems**.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. The **?** on each diagram shows what to find. Write the **formula**, **substitute** the numbers, then the **answer** with its unit.

3. Some Medium questions come from earlier lessons. The tag (like **10.2** or **Year 8**) says which one.

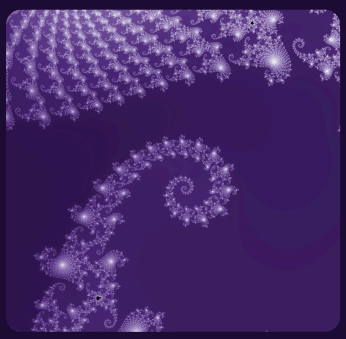
4. Stuck? Read the **Remember** line in the blue box, then look back at your notes and the example. Answers with π: **one decimal place** unless the question says otherwise.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
11.1	The length of an interval	3		☐
11.2	The midpoint of an interval	6		☐
11.3	The gradient of an interval	9		☐
11.4	Graphing linear equations	12		☐
11.5	The gradient–intercept equation	15		☐
11.6	Finding the equation of a line	18		☐
11.7	Solving linear equations graphically	21		☐
11.8	Linear modelling	24		☐
11.9	Graphing parabolas	27		☐
11.10	Parabolas in real life	30		☐

In this chapter you will

- find the length, midpoint and gradient of an interval on the number plane
- graph straight lines from a table of values, from intercepts and from $y = mx + c$
- find the equation of a straight line from its graph or from two points
- solve linear equations by reading a graph
- use straight-line graphs and equations to model real situations
- graph parabolas and use them in real situations



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is silver seahorse tails deep in Seahorse Valley, coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

11.1 The length of an interval

EASY

Learning intention We are learning to find the length of an interval on the number plane.

Success criteria
- ☐ I can use Pythagoras' theorem to find the length of an interval on a grid
- ☐ I can use the distance formula with any two points
- ☐ I can give a length as a whole number or correct to one decimal place

NOTES — Key ideas for this lesson

TEACHER COPY

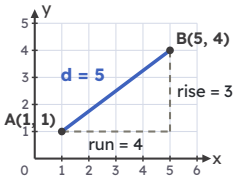
KEY TERMS

interval: part of a line between two endpoints
run: the change in x: $x_2 - x_1$
rise: the change in y: $y_2 - y_1$
length (d): the distance from one endpoint to the other
(x_1, y_1), (x_2, y_2): the two endpoints
√: square root

FORMULAS

$d^2 = \text{run}^2 + \text{rise}^2$
$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$
Horizontal: d = difference in x
Vertical: d = difference in y

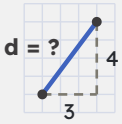
STEPS 1 find the run and the rise
2 square them and add 3 square root
4 round to 1 d.p.



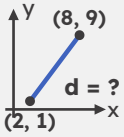
WE DO — With your teacher: copy what they write on the board.

Example 1 Find the length of each interval. Round to one decimal place where needed.

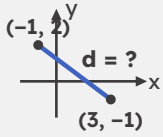
a Length



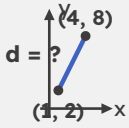
b Length



c Length



d Length



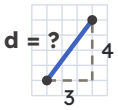
YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER run = across, rise = up or down · $d^2 = \text{run}^2 + \text{rise}^2$ (Pythagoras) · square root last · round to 1 d.p.

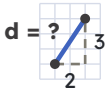
ROUND 1 — Questions 1–4: find the length of each interval.

1



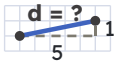
Formula $c^2 = a^2 + b^2$
Substitute _____
Answer _____

2



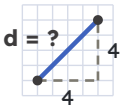
Formula $c^2 = a^2 + b^2$
Substitute _____
Answer _____

3



Formula $c^2 = a^2 + b^2$
Substitute _____
Answer _____

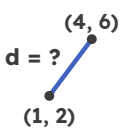
4



Formula $c^2 = a^2 + b^2$
Substitute _____
Answer _____

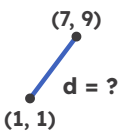
ROUND 2 — Questions 5–8: find the length of each interval.

5



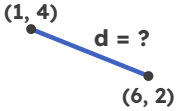
Formula _____
Substitute _____
Answer _____

6



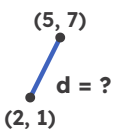
Formula _____
Substitute _____
Answer _____

7



Formula _____
Substitute _____
Answer _____

8



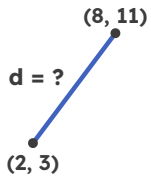
Formula _____
Substitute _____
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Subtract in the same order for x and y · a negative squared is positive.

ROUND 3 Questions 9–11: use the distance formula to find each length.

9

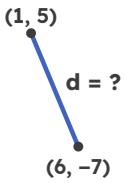


Formula $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$

Substitute

Answer

10

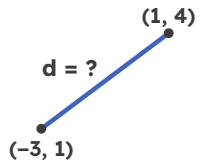


Formula $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$

Substitute

Answer

11



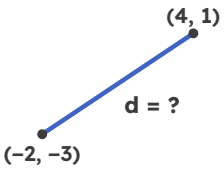
Formula $d^2 = (x_2-x_1)^2 + (y_2-y_1)^2$

Substitute

Answer

ROUND 4 Questions 12–14: find each length, correct to one decimal place.

12

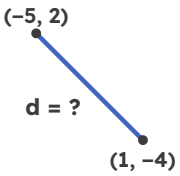


Formula

Substitute

Answer

13

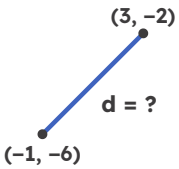


Formula

Substitute

Answer

14



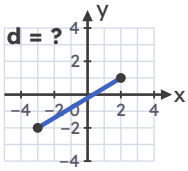
Formula

Substitute

Answer

ROUND 5 Questions 15–17: find the length of each interval on the grid.

15

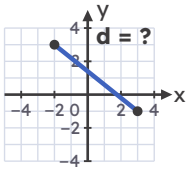


Formula

Substitute

Answer

16

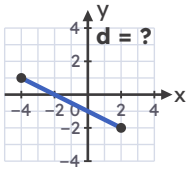


Formula

Substitute

Answer

17



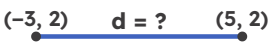
Formula

Substitute

Answer

ROUND 6 Questions 18–20: each interval is horizontal or vertical. Find its length.

18

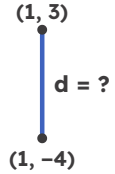


Formula

Substitute

Answer

19

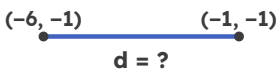


Formula

Substitute

Answer

20



Formula

Substitute

Answer

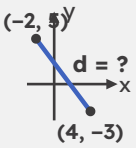
11.1 The length of an interval

MEDIUM

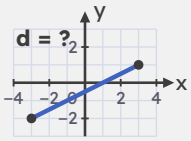
WE DO With your teacher: copy what they write on the board.

Example 2 Give lengths correct to one decimal place where needed.

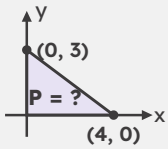
a Length



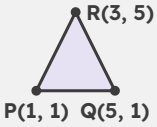
b Length



c Perimeter



d Isosceles?



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 Find the length of the interval joining (3, −2) and (9, 6).

Formula _____
Substitute _____
Answer _____

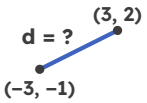
2 Find the hypotenuse of a right-angled triangle with shorter sides 9 cm and 12 cm. Ch 2

Formula _____
Substitute _____
Answer _____

3 Find the distance between A(−4, 1) and B(2, −7).

Formula _____
Substitute _____
Answer _____

4 Length



Formula _____
Substitute _____
Answer _____

5 Solve 3x − 7 = 14. Ch 7

Answer _____

6 Find the perimeter of the triangle with vertices (0, 0), (6, 0) and (6, 8).

Formula _____
Substitute _____
Answer _____

7 Which point is closer to the origin: A(5, −3) or B(−4, 4)?

Answer _____

8 Find the volume of a cylinder with radius 3 cm and height 10 cm, correct to one decimal place. 10.10

Formula _____
Substitute _____
Answer _____

9 The interval from (1, 2) to (k, 6) is 5 units long, and k > 1. Find k.

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 On a map, 1 unit = 100 m. A drone flies in a straight line from (−3, 2) to (5, 8). How far does it fly, in metres?

Answer _____

2 Town A is at (2, 1) and town B is at (14, 6) on a map grid in kilometres. The road goes 12 km east, then 5 km north. How much shorter is the straight-line distance than the road?

Answer _____

3 Show that the triangle with vertices A(1, 1), B(7, 1) and C(4, 5) is isosceles, then find its perimeter.

Answer _____

11.2 The midpoint of an interval

EASY

Learning intention We are learning to find the midpoint of an interval on the number plane.

Success criteria
- ☐ I can find the midpoint by averaging the x-values and the y-values
- ☐ I can find midpoints with negative and decimal coordinates
- ☐ I can find an endpoint when the midpoint and the other endpoint are known

NOTES — Key ideas for this lesson

TEACHER COPY

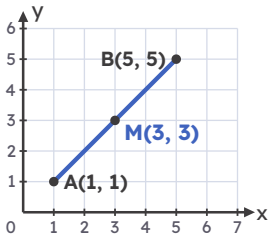
KEY TERMS

midpoint (M): the point halfway along an interval
average: add, then divide by how many
endpoints: A(x_1, y_1) and B(x_2, y_2)

FORMULAS

M = ((x_1 + x_2) ÷ 2, (y_1 + y_2) ÷ 2)
x of M = average of the x-values
y of M = average of the y-values
Endpoint: B = 2 × M − A (for x and y)

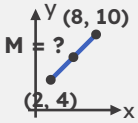
STEPS
1. add the x-values, ÷ 2
2. add the y-values, ÷ 2
3. write M as a point (x, y)



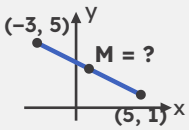
WE DO — With your teacher: copy what they write on the board.

Example 1 Find each midpoint, or the missing endpoint B.

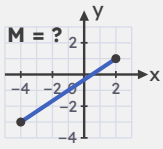
a Midpoint



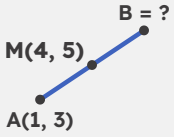
b Midpoint



c Midpoint



d Endpoint



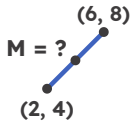
YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Add the two x-values and halve · add the two y-values and halve · write the answer as a point

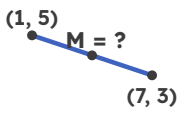
ROUND 1 — Questions 1–4: find the midpoint of each interval.

1



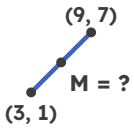
x (x₁+x₂)÷2
y
M

2



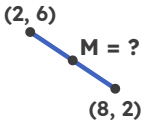
x (x₁+x₂)÷2
y
M

3



x (x₁+x₂)÷2
y
M

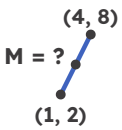
4



x (x₁+x₂)÷2
y
M

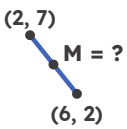
ROUND 2 — Questions 5–8: find each midpoint. The answers may be halves.

5



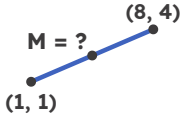
Formula
Substitute
Answer

6



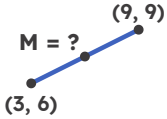
Formula
Substitute
Answer

7



Formula
Substitute
Answer

8



Formula
Substitute
Answer

11.2 The midpoint of an interval

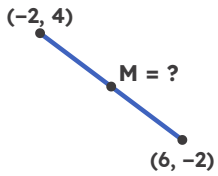
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Watch the signs: (−5 + 1) ÷ 2 = −2 · a midpoint can be a decimal.

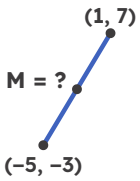
ROUND 3 Questions 9–11: use the midpoint formula.

9



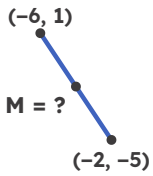
Formula _____
Substitute _____
Answer _____

10



Formula _____
Substitute _____
Answer _____

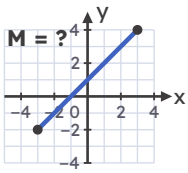
11



Formula _____
Substitute _____
Answer _____

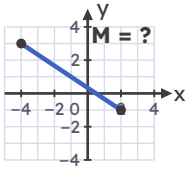
ROUND 4 Questions 12–14: find the midpoint of each interval on the grid.

12



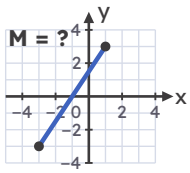
Formula _____
Substitute _____
Answer _____

13



Formula _____
Substitute _____
Answer _____

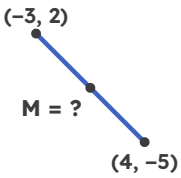
14



Formula _____
Substitute _____
Answer _____

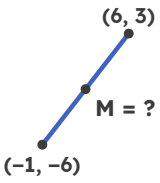
ROUND 5 Questions 15–17: find each midpoint. Give decimals where needed.

15



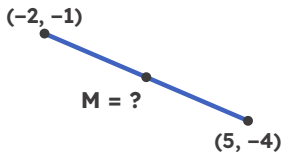
Formula _____
Substitute _____
Answer _____

16



Formula _____
Substitute _____
Answer _____

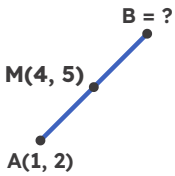
17



Formula _____
Substitute _____
Answer _____

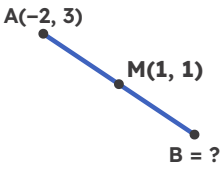
ROUND 6 Questions 18–20: M is the midpoint of AB. Find B.

18



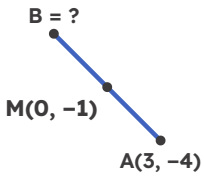
Formula _____
Substitute _____
Answer _____

19



Formula _____
Substitute _____
Answer _____

20



Formula _____
Substitute _____
Answer _____

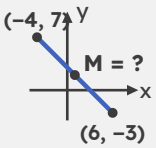
11.2 The midpoint of an interval

MEDIUM

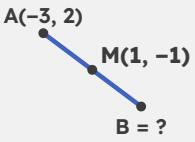
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each point.**

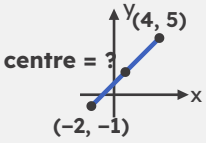
a Midpoint



b Endpoint



c Centre



d Find k if the midpoint of (k, 3) and (8, 9) is (5, 6).

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 Find the midpoint of the interval joining (−3, 8) and (7, −2).

Formula _____
Substitute _____
Answer _____

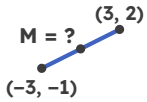
2 Find the length of the interval joining (1, 1) and (7, 9). [11.1]

Formula _____
Substitute _____
Answer _____

3 M is the midpoint of AB. A is (2, −5) and M is (6, −1). Find B.

Formula _____
Substitute _____
Answer _____

4 Midpoint



Formula _____
Substitute _____
Answer _____

5 Solve (x + 4) ÷ 2 = 7. [Ch 7]

Answer _____

6 The midpoint of (k, 3) and (8, 9) is (3, 6). Find k.

Answer _____

7 Find the midpoint of PQ, where P is (−6, −1) and Q is (−1, −8).

Formula _____
Substitute _____
Answer _____

8 Find the area of a circle with diameter 10 cm, correct to one decimal place. [10.7]

Formula _____
Substitute _____
Answer _____

9 A is (1, 2), B is (7, 2) and C is (7, 8). Find the midpoint of AC, then its distance from B.

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 Two friends live at (−4, 3) and (8, −1) on a town map (units in km). They meet halfway between their homes. Where do they meet?

Answer _____

2 A circle has a diameter with endpoints (−1, 2) and (7, 8). Find the centre and the radius of the circle.

Answer _____

3 A straight path runs from a gate G(2, 1) to a shed S(10, 7) (units in metres). A tap is halfway along the path. Find the tap's position and how far it is from the gate.

Answer _____

11.3 The gradient of an interval

EASY

Learning intention We are learning to find the gradient (steepness) of an interval or a line.

Success criteria
- ☐ I can say whether a gradient is positive, negative, zero or undefined
- ☐ I can find a gradient by counting rise and run on a grid
- ☐ I can use the gradient formula with any two points

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

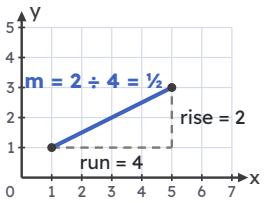
gradient (m): the steepness of a line: rise ÷ run
rise: change in y: up +, down –
run: change in x, left to right
positive: goes up to the right
negative: goes down to the right
zero: horizontal line
undefined: vertical line (run = 0)

FORMULAS

m = rise ÷ run
m = (y₂–y₁) ÷ (x₂–x₁)
Horizontal: m = 0
Vertical: m is undefined

STEPS

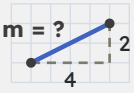
1. pick two points on the line
2. count the run (to the right)
3. count the rise (up + or down –)
4. m = rise ÷ run, simplified



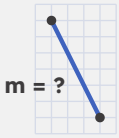
WE DO — With your teacher: copy what they write on the board.

Example 1 Find each gradient.

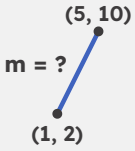
a Gradient



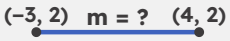
b Gradient



c Gradient



d Gradient



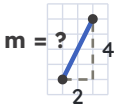
YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER m = rise ÷ run · up is +, down is – · always run to the right · simplify the fraction

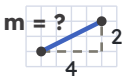
ROUND 1 — Questions 1–4: find the gradient of each interval.

1



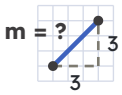
Formula: m = rise ÷ run
Substitute: _____
Answer: _____

2



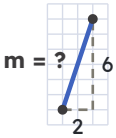
Formula: m = rise ÷ run
Substitute: _____
Answer: _____

3



Formula: m = rise ÷ run
Substitute: _____
Answer: _____

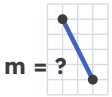
4



Formula: m = rise ÷ run
Substitute: _____
Answer: _____

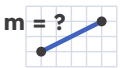
ROUND 2 — Questions 5–8: count the rise and run, then find each gradient.

5



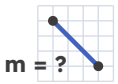
Formula: _____
Substitute: _____
Answer: _____

6



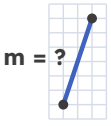
Formula: _____
Substitute: _____
Answer: _____

7



Formula: _____
Substitute: _____
Answer: _____

8



Formula: _____
Substitute: _____
Answer: _____

11.3 The gradient of an interval

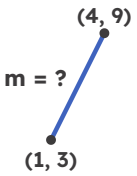
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER $m = (y_2 - y_1) \div (x_2 - x_1)$ · subtract in the same order · a negative ÷ a positive is negative

ROUND 3 Questions 9–11: use the gradient formula.

9

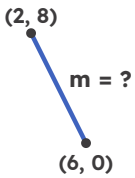


Formula $m = (y_2 - y_1) \div (x_2 - x_1)$

Substitute

Answer

10

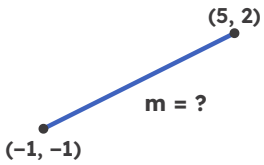


Formula $m = (y_2 - y_1) \div (x_2 - x_1)$

Substitute

Answer

11



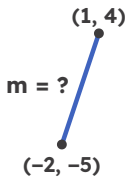
Formula $m = (y_2 - y_1) \div (x_2 - x_1)$

Substitute

Answer

ROUND 4 Questions 12–14: find each gradient. Simplify fractions.

12

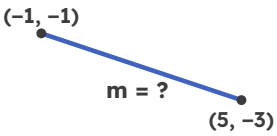


Formula

Substitute

Answer

13

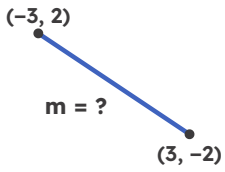


Formula

Substitute

Answer

14



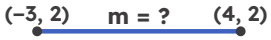
Formula

Substitute

Answer

ROUND 5 Questions 15–17: is the gradient positive, negative, zero or undefined? Find it if you can.

15

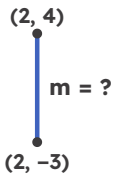


Type

Working

Gradient

16

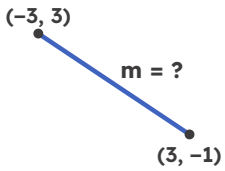


Type

Working

Gradient

17



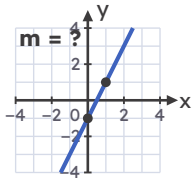
Type

Working

Gradient

ROUND 6 Questions 18–20: find the gradient of each line.

18

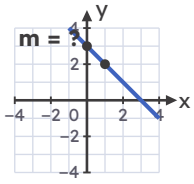


Formula

Substitute

Answer

19

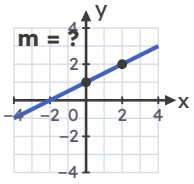


Formula

Substitute

Answer

20



Formula

Substitute

Answer

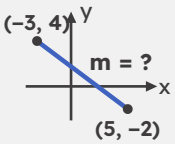
11.3 The gradient of an interval

MEDIUM

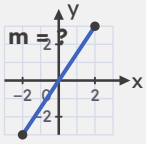
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each gradient or value.**

a Gradient



b Gradient



c The line through (1, 2) and (4, k) has gradient 3. Find k.

d Which line is steeper: m = −3 or m = 2?

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 Find the gradient of the interval joining (2, −1) and (6, 7).

Formula _____
Substitute _____
Answer _____

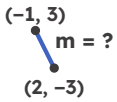
2 Find the midpoint of (−2, 5) and (6, 1). 11.2

Formula _____
Substitute _____
Answer _____

3 Find the gradient of the line through (−4, 3) and (2, −6).

Formula _____
Substitute _____
Answer _____

4 Gradient



Formula _____
Substitute _____
Answer _____

5 Find the length of the interval from (−1, 2) to (5, 10). 11.1

Formula _____
Substitute _____
Answer _____

6 Is the gradient of the line through (3, 5) and (3, −2) zero or undefined? Explain.

Answer _____

7 A ramp rises 1.2 m over a run of 4.8 m. Find its gradient.

Formula _____
Substitute _____
Answer _____

8 Find the volume of a rectangular prism 8 cm long, 5 cm wide and 3 cm high. 10.10

Formula _____
Substitute _____
Answer _____

9 The line through (2, k) and (5, 11) has gradient 3. Find k.

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A road climbs 45 m over a horizontal distance of 600 m. Find its gradient as a decimal and as a percentage.

Answer _____

2 A roof rises 2.4 m over a horizontal distance of 4 m. A builder says the roof is steeper than a ramp with gradient ½. Is the builder right? Explain.

Answer _____

3 Show that the points A(0, 1), B(2, 5) and C(5, 11) lie on one straight line.

Answer _____

11.4 Graphing linear equations

EASY

Learning intention We are learning to graph straight lines and find where they cross the axes.

Success criteria
- ☐ I can complete a table of values and graph a straight line
- ☐ I can find the x- and y-intercepts of a line
- ☐ I can test whether a point lies on a line, and graph lines like x = 3 and y = −2

NOTES — Key ideas for this lesson

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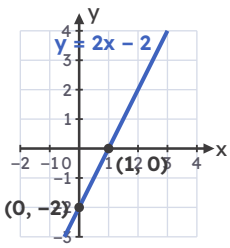
KEY TERMS

linear equation: its graph is a straight line
table of values: x-values and the y-values they give
x-intercept: where a line crosses the x-axis (y = 0)
y-intercept: where a line crosses the y-axis (x = 0)
x = a: a vertical line
y = b: a horizontal line

FORMULAS

x-intercept: put y = 0 and solve
y-intercept: put x = 0
On the line? Substitute x and y: both sides equal

STEPS
1. make a table of values
2. substitute each x to find y
3. plot the points
4. rule a line through them



WE DO — With your teacher: copy what they write on the board.

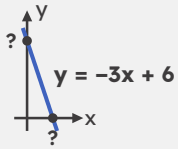
Example 1 Complete, find or graph as shown.

a Table

y = 2x + 1

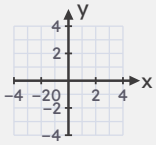
x	−1	0	1	2
y				

b Intercepts



c Does (4, −2) lie on the line y = x − 6?

d x = 3 and y = −2



YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Substitute each x into the rule · y-intercept: x = 0 · x-intercept: y = 0

ROUND 1 — Questions 1–4: complete each table of values.

1

x	0	1	2	3
y				

Rule y = 2x + 3
Working
y-values

2

x	0	1	2	3
y				

Rule y = x − 2
Working
y-values

3

x	0	1	2	3
y				

Rule y = 3x
Working
y-values

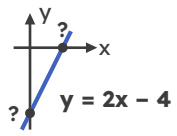
4

x	0	1	2	3
y				

Rule y = −x + 4
Working
y-values

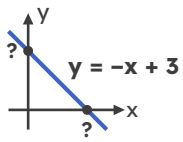
ROUND 2 — Questions 5–8: find the x- and y-intercepts of each line.

5



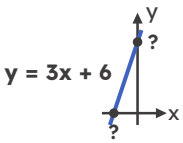
y-intercept
x-intercept
Answer

6



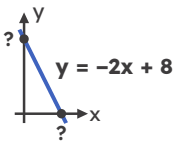
y-intercept
x-intercept
Answer

7



y-intercept
x-intercept
Answer

8



y-intercept
x-intercept
Answer

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER A negative x: use brackets, 2 × (−1) − 1 = −3 · an x-intercept can be a fraction.

ROUND 3 Questions 9–11: complete each table of values.

9 $y = 2x - 1$

x	−1	0	1	2
y				

Substitute _____
Working _____
y-values _____

10 $y = -2x + 3$

x	−1	0	1	2
y				

Substitute _____
Working _____
y-values _____

11 $y = \frac{1}{2}x + 2$

x	−2	0	2	4
y				

Substitute _____
Working _____
y-values _____

ROUND 4 Questions 12–14: test whether the point lies on the line.

12 Is (3, 5) on $y = 2x - 1$?

Substitute _____
Working _____
Yes or no _____

13 Is (−2, 4) on $y = -3x - 1$?

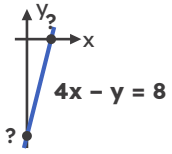
Substitute _____
Working _____
Yes or no _____

14 Is (1, −3) on $y = 4x - 7$?

Substitute _____
Working _____
Yes or no _____

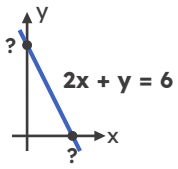
ROUND 5 Questions 15–17: find both intercepts of each line.

15



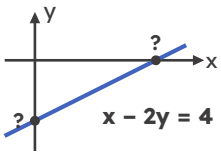
y-intercept _____
x-intercept _____
Answer _____

16



y-intercept _____
x-intercept _____
Answer _____

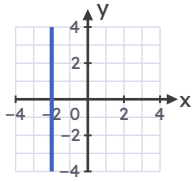
17



y-intercept _____
x-intercept _____
Answer _____

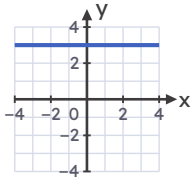
ROUND 6 Questions 18–20: write the equation of each line.

18



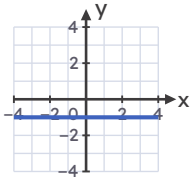
Type _____
A point _____
Equation _____

19



Type _____
A point _____
Equation _____

20



Type _____
A point _____
Equation _____

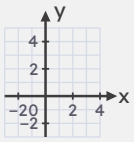
11.4 Graphing linear equations

MEDIUM

WE DO With your teacher: copy what they write on the board.

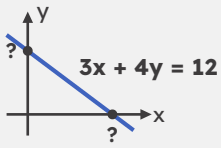
Example 2 **Graph, test or find as shown.**

a Graph $y = -2x + 4$

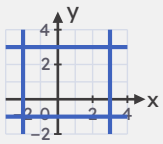


b Does (4, −2) lie on the line $x - 3y = 10$?

c Intercepts



d Area



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 Find the x- and y-intercepts of $y = 2x + 10$.

Formula _____
Substitute _____
Answer _____

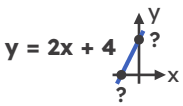
2 Find the gradient of the line through (−1, 4) and (3, −4). [11.3]

Formula _____
Substitute _____
Answer _____

3 Does the point (−3, 7) lie on $y = -2x + 1$? Show your working.

Answer _____

4 Intercepts



Answer _____

5 Find the length of the interval joining (−2, −1) and (4, 7). [11.1]

Formula _____
Substitute _____
Answer _____

6 Write the equation of the horizontal line through (5, −3).

Answer _____

7 Find the x- and y-intercepts of $5x - 2y = 20$.

Formula _____
Substitute _____
Answer _____

8 Find the midpoint of (−7, 2) and (3, −6). [11.2]

Formula _____
Substitute _____
Answer _____

9 Find the area of the triangle between the axes and the line $y = -2x + 6$.

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A plumber charges $C = 40 + 60h$ dollars for h hours of work. Make a table for $h = 0$, 1, 2 and 3. What is the C-intercept, and what does it mean?

Answer _____

2 The line $3x + 2y = 18$ crosses both axes. Find where, then find the length of the interval between these two points, correct to one decimal place.

Answer _____

3 The lines $x = -1$, $x = 5$, $y = 2$ and $y = -3$ enclose a rectangle. Find its perimeter and its area.

Answer _____

11.5 The gradient–intercept equation

EASY

Learning intention We are learning to use y = mx + c to find the gradient and y-intercept of a line, and to graph it.

Success criteria
- ☐ I can state the gradient and y-intercept of a line in the form y = mx + c
- ☐ I can rearrange an equation into the form y = mx + c
- ☐ I can graph a line from its gradient and y-intercept

NOTES — Key ideas for this lesson

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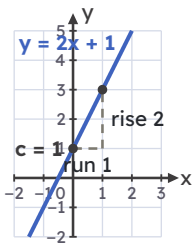
KEY TERMS

y = mx + c: the gradient–intercept form
m: the gradient: the number in front of x
c: the y-intercept: the constant term
parallel lines: lines with the same gradient
subject: the variable on its own on one side (y = …)

FORMULAS

y = mx + c
m = gradient, c = y-intercept
Parallel lines: equal gradients

STEPS (1) make y the subject (2) read m and c (3) plot (0, c) (4) step run 1, rise m; rule the line



WE DO — With your teacher: copy what they write on the board.

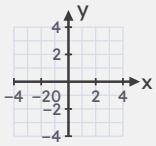
Example 1 State m and c, or graph the line.

a y = 3x + 1

b y = 8 − x

c y = ⅕x + 4

d Graph y = 2x − 3



YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER m is the number in front of x (with its sign) · c is the number on its own · y must be on its own first

ROUND 1 — Questions 1–4: state the gradient and the y-intercept.

1 y = 4x + 3
Formula y = mx + c
m = _____
c = _____

2 y = −2x + 5
Formula y = mx + c
m = _____
c = _____

3 y = x − 7
Formula y = mx + c
m = _____
c = _____

4 y = ½x + 1
Formula y = mx + c
m = _____
c = _____

ROUND 2 — Questions 5–8: state the gradient and the y-intercept.

5 y = 6 − 3x
Rearrange _____
m = _____
c = _____

6 y = −x
Rearrange _____
m = _____
c = _____

7 y = 9
Rearrange _____
m = _____
c = _____

8 y = 0.5x − 2
Rearrange _____
m = _____
c = _____

11.5 The gradient–intercept equation

EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Move the x-term first, then divide every term · c is where you start; m is up (or down) for each 1 across.

ROUND 3 Questions 9–11: rearrange into the form y = mx + c, then state m and c.

9 2x + y = 7

Formula y = mx + c

Rearrange

m and c

10 y − 3x = 4

Formula y = mx + c

Rearrange

m and c

11 4x + 2y = 10

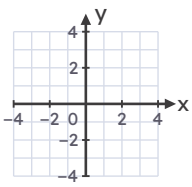
Formula y = mx + c

Rearrange

m and c

ROUND 4 Questions 12–14: graph each line from its gradient and y-intercept.

12 y = 2x − 1

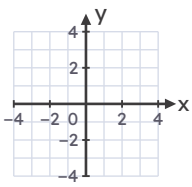


Start (0, c)

Step

Two points

13 y = −x + 3

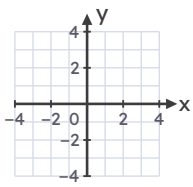


Start (0, c)

Step

Two points

14 y = ½x + 1



Start (0, c)

Step

Two points

ROUND 5 Questions 15–17: write the equation of each line.

15 m = 3, c = −2

Formula

Substitute

Answer

16 m = −½, c = 4

Formula

Substitute

Answer

17 m = 0, c = 5

Formula

Substitute

Answer

ROUND 6 Questions 18–20: are the two lines parallel?

18 y = 2x + 1 and y = 2x − 5

m of each

Compare

Answer

19 y = 3x + 2 and y = −3x + 2

m of each

Compare

Answer

20 y = 4 − x and y = −x + 7

m of each

Compare

Answer

11.5 The gradient–intercept equation

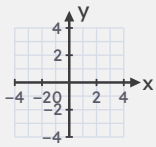
MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Find, graph or write as shown.**

a State m and c for $3x + 2y = 12$.

b Graph $y = -2x + 4$



c Write the equation of the line with gradient −3 and y-intercept 5.

d Write the equation of the line parallel to $y = 4x - 1$ with y-intercept 3.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 State the gradient and y-intercept of $y = -6x$.

Formula _____
Substitute _____
Answer _____

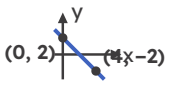
2 Find the x-intercept of $y = 3x - 12$. 11.4

Formula _____
Substitute _____
Answer _____

3 Rearrange $6x + 3y = 9$ into the form $y = mx + c$.

Formula _____
Substitute _____
Answer _____

4 Equation



Formula _____
Substitute _____
Answer _____

5 Find the gradient of the interval joining (2, 3) and (6, 11). 11.3

Formula _____
Substitute _____
Answer _____

6 Write the equation of the line with gradient ¾ and y-intercept −2.

Formula _____
Substitute _____
Answer _____

7 Are $y = 2x + 7$ and $4x - 2y = 6$ parallel? Show your working.

Answer _____

8 Find the midpoint of (−3, −4) and (5, 10). 11.2

Formula _____
Substitute _____
Answer _____

9 A line is parallel to $y = -3x + 1$ and crosses the y-axis at −4. Write its equation.

Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A phone plan costs $C = 15 + 0.5g$ dollars a month for g gigabytes of data. State the gradient and the C-intercept, and say what each means.

Answer _____

2 Line A has equation $y = 2x + 3$. Line B is parallel to line A and crosses the y-axis 5 units below it. Write the equation of line B, then find its x-intercept.

Answer _____

3 A student rewrote $2y - 6x = 8$ as $y = 6x + 8$. Find the mistake, write the correct equation and state m and c.

Answer _____

11.6 Finding the equation of a line

EASY

Learning intention We are learning to find the equation of a straight line from its graph or from two points on it.

Success criteria
- ☐ I can read the y-intercept and gradient from a graph
- ☐ I can write the equation of a line as y = mx + c
- ☐ I can find the equation of a line through two points

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

y-intercept (c): where the line crosses the y-axis
gradient (m): rise ÷ run between two points
y = mx + c: the equation of the line

FORMULAS

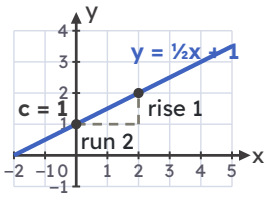
y = mx + c

m = rise ÷ run = $(y_2 - y_1) \div (x_2 - x_1)$

Two points: find m, then substitute a point to find c

STEPS

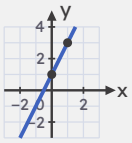
1. find c where the line crosses the y-axis
2. find m from two points
3. write y = mx + c
4. check with another point



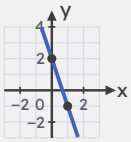
WE DO — With your teacher: copy what they write on the board.

Example 1 Find the equation of each line.

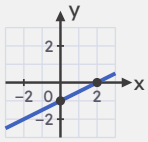
a Equation



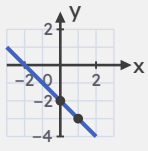
b Equation



c Equation



d Equation



YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER c: where the line crosses the y-axis · m = rise ÷ run between the two points · then write y = mx + c

ROUND 1 — Questions 1–4: find the equation of each line.

1 (0, 2) (3, 5)

Formula y = mx + c
Substitute
Answer

2 (0, −1) (3, 5)

Formula y = mx + c
Substitute
Answer

3 (0, 1) (3, −2)

Formula y = mx + c
Substitute
Answer

4 (0, 0) (3, 9)

Formula y = mx + c
Substitute
Answer

ROUND 2 — Questions 5–8: find the equation of each line.

5 (0, 3) (3, −3)

Formula
Substitute
Answer

6 (0, 1) (2, 2)

Formula
Substitute
Answer

7 (0, −1) (3, −10)

Formula
Substitute
Answer

8 (0, −3) (3, 0)

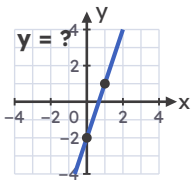
Formula
Substitute
Answer

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER No point on the y-axis? Find m, then substitute one point into y = mx + c to find c.

ROUND 3 Questions 9–11: find the equation of each line.

9

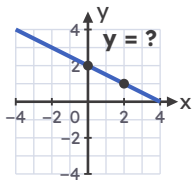


Formula y = mx + c

Substitute

Answer

10

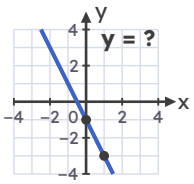


Formula y = mx + c

Substitute

Answer

11



Formula y = mx + c

Substitute

Answer

ROUND 4 Questions 12–14: find the equation of the line through the two points.

12 (0, 3) and (2, 7)

Formula

Substitute

Answer

13 (0, −1) and (4, 1)

Formula

Substitute

Answer

14 (0, 4) and (2, −2)

Formula

Substitute

Answer

ROUND 5 Questions 15–17: find the equation of the line through the two points.

15 (1, 5) and (3, 9)

m

c

Equation

16 (2, 1) and (4, 7)

m

c

Equation

17 (−2, 6) and (2, −2)

m

c

Equation

ROUND 6 Questions 18–20: find the equation from the gradient and a point.

18 m = 3, through (2, 5)

m

c

Equation

19 m = −2, through (1, 4)

m

c

Equation

20 m = ½, through (4, 1)

m

c

Equation

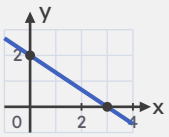
11.6 Finding the equation of a line

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Find each equation.**

a Equation



b Find the equation of the line through (−1, 7) and (2, −2).

c A line has gradient 4 and passes through (−1, −6). Find its equation.

d Does (3, 10) lie on the line in part c?

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 Find the equation of the line through (0, −5) with gradient 4.

Formula _____
Substitute _____
Answer _____

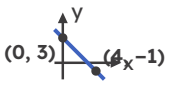
2 State m and c for y = 7 − 2x. 11.5

Formula _____
Substitute _____
Answer _____

3 Find the equation of the line through (1, 1) and (3, 7).

Formula _____
Substitute _____
Answer _____

4 Equation



Formula _____
Substitute _____
Answer _____

5 Find the y-intercept of 2x + 5y = 20. 11.4

Formula _____
Substitute _____
Answer _____

6 Find the equation of the line through (−2, 3) and (4, 6).

Formula _____
Substitute _____
Answer _____

7 A line with gradient −3 passes through (2, −1). Find its equation.

Formula _____
Substitute _____
Answer _____

8 Find the distance between (−3, 0) and (3, 8). 11.1

Formula _____
Substitute _____
Answer _____

9 Find the equation of the line through (−1, 4) that is parallel to y = 2x − 7.

Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A candle is 24 cm tall when it is lit, and 18 cm tall after 2 hours. It burns at a steady rate. Write an equation for its height h cm after t hours, and find when it burns out.

Answer _____

2 A 4 km taxi trip costs $13 and a 10 km trip costs $25. Find an equation for the cost C dollars of a d km trip. What is the flagfall (the charge for 0 km)?

Answer _____

3 Find the equation of the line through A(−2, −1) and B(4, 11), then find where it crosses the x-axis.

Answer _____

11.7 Solving linear equations graphically

EASY

Learning intention We are learning to solve linear equations by reading a straight-line graph.

Success criteria
- ☐ I can use the graph of y = mx + c to solve mx + c = k
- ☐ I can find the point where two lines cross
- ☐ I can check a graphical solution by substituting

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

solution: the value of x that makes the equation true
point of intersection: where two lines cross
y = k: a horizontal line at height k

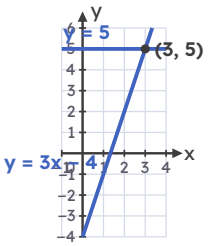
FORMULAS

Solve 3x − 4 = 5: graph y = 3x − 4 and y = 5; read x where they meet

Two lines: the solution is the crossing point (x, y)

Check: substitute x into the equation

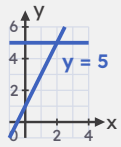
STEPS
1. graph y = mx + c
2. draw the line y = k
3. read x at the crossing
4. check by substituting



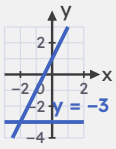
WE DO — With your teacher: copy what they write on the board.

Example 1 Use the graphs to solve.

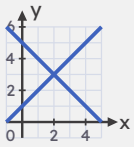
a 2x + 1 = 5



b 2x + 1 = −3



c Crossing point



d Check part a by substituting.

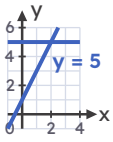
YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Find the height k on the y-axis · go across to the line · go down to read x

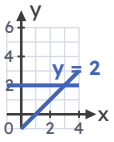
ROUND 1 — Questions 1–4: use each graph to solve the equation.

1



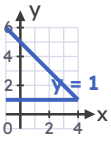
Equation 2x + 1 = 5
Read x
Answer

2



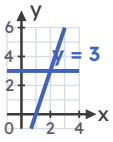
Equation x − 1 = 2
Read x
Answer

3



Equation −x + 5 = 1
Read x
Answer

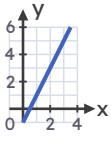
4



Equation 3x − 3 = 3
Read x
Answer

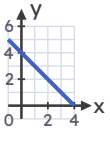
ROUND 2 — Questions 5–8: draw the line y = k on each graph, then solve.

5 2x − 1 = 3



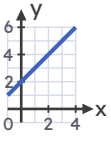
Draw y =
Read x
Answer

6 −x + 4 = 1



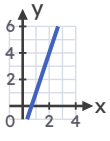
Draw y =
Read x
Answer

7 x + 2 = 5



Draw y =
Read x
Answer

8 3x − 2 = 4



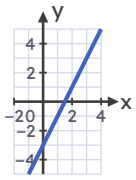
Draw y =
Read x
Answer

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Read the x-value straight down (or up) from the crossing · two lines: read both x and y.

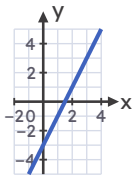
ROUND 3 Questions 9–11: use the graph of y = 2x − 3 to solve each equation.

9



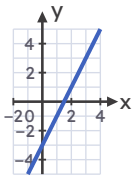
Equation 2x − 3 = 1
Read x
Answer

10



Equation 2x − 3 = −5
Read x
Answer

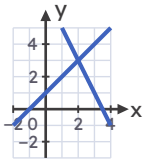
11



Equation 2x − 3 = 3
Read x
Answer

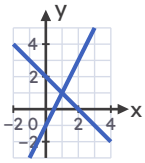
ROUND 4 Questions 12–14: find the point where the two lines cross.

12



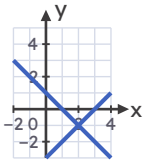
Read x
Read y
Point

13



Read x
Read y
Point

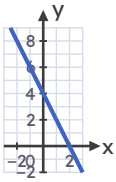
14



Read x
Read y
Point

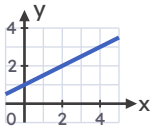
ROUND 5 Questions 15–17: solve from the graph, then check by substituting.

15 −2x + 4 = 8



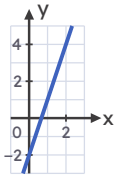
Read x
Check
Answer

16 ½x + 1 = 3



Read x
Check
Answer

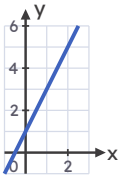
17 3x − 2 = 4



Read x
Check
Answer

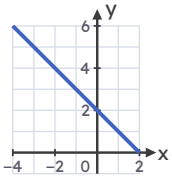
ROUND 6 Questions 18–20: the answers are halves. Solve from the graph.

18 2x + 1 = 4



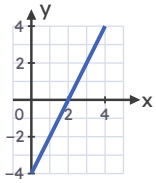
Read x
Check
Answer

19 −x + 2 = 4.5



Read x
Check
Answer

20 2x − 4 = 3



Read x
Check
Answer

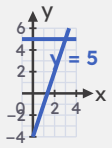
11.7 Solving linear equations graphically

MEDIUM

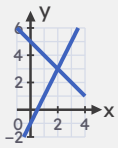
WE DO With your teacher: copy what they write on the board.

Example 2 **Solve using graphs, then check.**

a Solve $3x - 4 = 5$



b Crossing point



c Use the graphs of $y = x + 3$ and $y = -2x + 6$ to solve $x + 3 = -2x + 6$.

d Check your answer to part b in both equations.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 The graph of $y = -2x + 1$ passes through $(-2, 5)$. Use this to solve $-2x + 1 = 5$.

Answer _____

2 Find the equation of the line through $(0, 2)$ and $(3, 8)$. 11.6

Formula _____

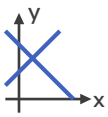
Substitute _____

Answer _____

3 Solve $4x - 3 = 9$ by algebra.

Answer _____

4 Crossing point



Answer _____

5 Find the x- and y-intercepts of $y = -x + 6$. 11.4

Formula _____

Substitute _____

Answer _____

6 The lines $y = 2x + 1$ and $y = 7$ cross. Find the point without drawing.

Answer _____

7 Where do $y = 3x - 2$ and $y = x + 4$ meet? Use a table of values for $x = 0$ to 4.

Answer _____

8 Write the equation of the line parallel to $y = -x + 2$ with y-intercept 6. 11.5

Formula _____

Substitute _____

Answer _____

9 Sketch $y = 0.5x + 2$, then use it to solve $0.5x + 2 = 4.5$.

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 Plan A costs $C = 20 + 5n$ dollars and plan B costs $C = 10 + 7n$ dollars for n lessons. Graph both for $n = 0$ to 8. After how many lessons do they cost the same, and how much is it?

Answer _____

2 A tank holds 40 L and drains at 5 L per minute, so $V = 40 - 5t$. Graph V for $t = 0$ to 8 and use it to find when the tank holds 15 L.

Answer _____

3 Sam has \$30 and saves \$12 a week. Mia has \$90 and spends \$8 a week. Graph both amounts. After how many weeks do they have the same amount, and how much is it?

Answer _____

11.8 Linear modelling

EASY

Learning intention We are learning to use straight-line graphs and equations to model real situations.

Success criteria
- ☐ I can use a rule to find a value, and find a rule from a table
- ☐ I can explain what the gradient and y-intercept mean in a real situation
- ☐ I can write and use an equation to make predictions

NOTES — Key ideas for this lesson

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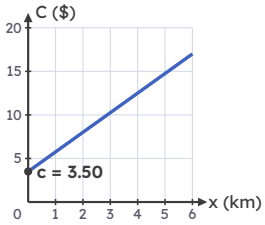
KEY TERMS

model: an equation or graph that describes a real situation
rate (gradient): the change for each 1 unit, e.g. $2.25 per km
starting value (c): the value when x = 0, e.g. a flagfall
predict: use the model to find a value

FORMULAS

y = mx + c
m = rate (change per unit)
c = starting value (x = 0)

STEPS 1 find c: the value at the start
2 find m: the change for each 1
3 write y = mx + c 4 substitute to predict



WE DO — With your teacher: copy what they write on the board.

Example 1 A rideshare costs $3.50 flagfall plus $2.25 for each kilometre.

a Table

x (km)	0	1	2	3
C ($)				

b What do the gradient and the C-intercept mean?

c Write an equation for the cost C of an x km trip.

d Find the cost of a 12 km trip.

YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Substitute the value into the rule · c is the starting value · m is the change for each 1

ROUND 1 — Questions 1–4: use each rule to find the value.

1 C = 5 + 2n, when n = 8
Rule: C = 5 + 2n
Substitute: _____
Answer: _____

2 C = 12 + 3h, when h = 5
Rule: C = 12 + 3h
Substitute: _____
Answer: _____

3 d = 60t, when t = 2.5
Rule: d = 60t
Substitute: _____
Answer: _____

4 V = 50 − 4t, when t = 6
Rule: V = 50 − 4t
Substitute: _____
Answer: _____

ROUND 2 — Questions 5–8: find the rule for each table.

5

x	0	1	2	3
y	5	8	11	14

Formula: _____
Substitute: _____
Answer: _____

6

x	0	1	2	3
y	2	6	10	14

Formula: _____
Substitute: _____
Answer: _____

7

x	0	1	2	3
y	20	17	14	11

Formula: _____
Substitute: _____
Answer: _____

8

x	0	1	2	3
y	1	1.5	2	2.5

Formula: _____
Substitute: _____
Answer: _____

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Starting value → c · amount per unit → m · read across, then up (or down), to use a graph.

ROUND 3 Questions 9–11: write a rule y = mx + c for each situation.

9 Taxi: $4 flagfall, then $2 per km

Formula y = mx + c
Substitute
Answer

10 Phone: $25 a month, plus 10c per text

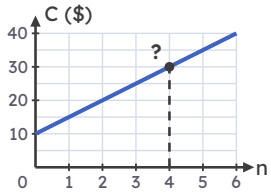
Formula y = mx + c
Substitute
Answer

11 Pool: holds 200 L, filling at 15 L a minute

Formula y = mx + c
Substitute
Answer

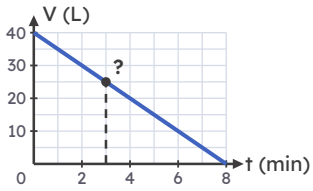
ROUND 4 Questions 12–14: read the value marked ? on each graph, then check it with a rule.

12



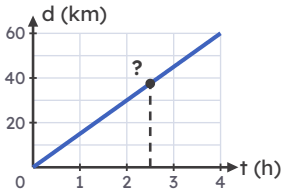
Read
Rule
Check

13



Read
Rule
Check

14



Read
Rule
Check

ROUND 5 Questions 15–17: say what m and c mean in each model.

15 C = 50 + 30h (a plumber, h hours)

m means
c means
Answer

16 H = 150 + 6w (a plant, w weeks)

m means
c means
Answer

17 B = 400 − 25w (a bank balance, w weeks)

m means
c means
Answer

ROUND 6 Questions 18–20: use the rule to find the missing value.

18 C = 4 + 2d. Find d when C = 30

Equation
Solve
Answer

19 C = 25 + 0.1t. Find t when C = 32

Equation
Solve
Answer

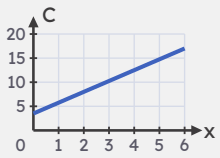
20 V = 200 + 15t. Find t when V = 500

Equation
Solve
Answer

WE DO With your teacher: copy what they write on the board.

Example 2 **Use the rideshare model C = 2.25x + 3.5 (from Example 1).**

a Read: cost of 3.5 km?



b Find the cost of a 76 km trip.

c A trip cost $12.05. How far was it?

d A trip cost $82.25. How far was it?

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 A plumber charges $80 plus $60 per hour. Write a rule for the cost C of h hours.

Formula _____
Substitute _____
Answer _____

2 Find the equation of the line through (0, 4) and (2, 10). 11.6

Formula _____
Substitute _____
Answer _____

3 Use your rule from question 1 to find the cost of 3.5 hours.

Formula _____
Substitute _____
Answer _____

4 Rule

x	0	1	2	3
y	12	10	8	6

Formula _____
Substitute _____
Answer _____

5 Where do y = x + 1 and y = 5 − x cross? 11.7

Answer _____

6 A car's tank holds 60 L of fuel. The car uses 7.5 L per 100 km. Write a rule for the fuel F left after d hundred kilometres.

Formula _____
Substitute _____
Answer _____

7 Using question 6, how far can the car go before the tank is empty?

Formula _____
Substitute _____
Answer _____

8 A line rises 6 for a run of 4. Find its gradient. 11.3

Formula _____
Substitute _____
Answer _____

9 Gym plan A: $50 + $10 a week. Plan B: $20 + $15 a week. After how many weeks do they cost the same?

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A candle is 30 cm tall and burns 2.5 cm each hour. Write a rule for its height h cm after t hours. Find its height after 5 hours, and when it burns out.

Answer _____

2 A mobile plan costs $20 a month plus 15c for each minute of calls. Write a rule for the monthly cost C of m minutes. Find the cost of 120 minutes, and how many minutes give a $50 bill.

Answer _____

3 A pool already holds 1200 L of water. Water is pumped in at 80 L a minute. Write a rule for the volume V after t minutes, and find how long until it holds 5000 L.

Answer _____

11.9 Graphing parabolas

EASY

Learning intention We are learning to graph parabolas of the form $y = ax^2 + c$ and describe their features.

Success criteria
- ☐ I can complete a table of values and graph a parabola
- ☐ I can find the vertex and say whether a parabola is concave up or concave down
- ☐ I can describe how a and c change the shape and position of a parabola

NOTES — Key ideas for this lesson

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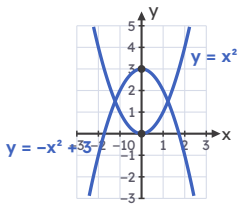
KEY TERMS

parabola: the U-shaped graph of $y = ax^2 + c$
vertex: the turning point: (0, c) here
concave up: opens upward (∪): $a > 0$
concave down: opens downward (∩): $a < 0$
axis of symmetry: the line $x = 0$: each side is a mirror image
wider / narrower: a closer to 0 is wider

FORMULAS

$y = ax^2 + c$
vertex (0, c)
$a > 0$: concave up; $a < 0$: concave down
bigger a (ignoring the sign): narrower

STEPS
1. make a table for $x = -2$ to 2
2. square x first, then × a, then + c
3. plot the points
4. join them with a smooth curve



WE DO — With your teacher: copy what they write on the board.

Example 1 Complete, describe or match each parabola.

a Table

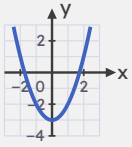
$y = x^2 + 1$

x	−2	−1	0	1	2
y					

b Find the vertex of $y = -2x^2$. Is it concave up or down?

c Which is wider: $y = x^2$ or $y = 3x^2$?

d Equation



YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Square x first: $(-2)^2 = 4$ · then multiply by a · then add c

ROUND 1 — Questions 1–4: complete each table of values.

1

x	−2	−1	0	1	2
y					

Rule $y = x^2$
Working
y-values

2

x	−2	−1	0	1	2
y					

Rule $y = x^2 + 2$
Working
y-values

3

x	−2	−1	0	1	2
y					

Rule $y = 2x^2$
Working
y-values

4

x	−2	−1	0	1	2
y					

Rule $y = -x^2$
Working
y-values

ROUND 2 — Questions 5–8: complete each table of values.

5 $y = x^2 - 3$

x	−2	−1	0	1	2
y					

Square x
Working
y-values

6 $y = -x^2 + 4$

x	−2	−1	0	1	2
y					

Square x
Working
y-values

7 $y = 3x^2$

x	−2	−1	0	1	2
y					

Square x
Working
y-values

8 $y = -2x^2 + 1$

x	−2	−1	0	1	2
y					

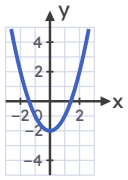
Square x
Working
y-values

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Vertex (0, c) · a > 0: ∪ · a < 0: ∩ · the a closer to 0 gives the wider parabola

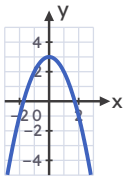
ROUND 3 Questions 9–11: state the vertex, and whether the parabola is concave up or down.

9



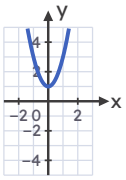
Vertex _____
Concave _____
Answer _____

10



Vertex _____
Concave _____
Answer _____

11



Vertex _____
Concave _____
Answer _____

ROUND 4 Questions 12–14: state the vertex and the concavity from the equation.

12 $y = -4x^2$

Vertex _____
Concave _____
Answer _____

13 $y = 0.5x^2 - 6$

Vertex _____
Concave _____
Answer _____

14 $y = 7 - x^2$

Vertex _____
Concave _____
Answer _____

ROUND 5 Questions 15–17: which parabola is wider? Give a reason.

15 $y = 2x^2$ or $y = 5x^2$

Compare a _____
Reason _____
Answer _____

16 $y = -x^2$ or $y = -\frac{1}{3}x^2$

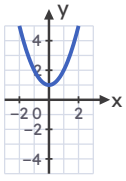
Compare a _____
Reason _____
Answer _____

17 $y = 4x^2 + 1$ or $y = x^2 + 4$

Compare a _____
Reason _____
Answer _____

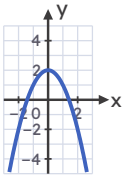
ROUND 6 Questions 18–20: write the equation of each parabola.

18



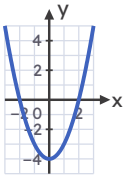
Vertex _____
Concave _____
Equation _____

19



Vertex _____
Concave _____
Equation _____

20

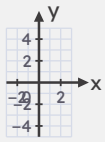


Vertex _____
Concave _____
Equation _____

WE DO With your teacher: copy what they write on the board.

Example 2 **Graph, choose or find as shown.**

a Graph $y = -x^2 + 4$



b Which is concave up with vertex (0, 0)? $y = x^2 - 1$, $y = 2x^2$, $y = -x^2$, $y = 3x^2 + 2$

c Does (3, 10) lie on $y = x^2 + 1$?

d Find the x-intercepts of $y = x^2 - 9$.

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 Find the vertex of $y = -3x^2 + 5$ and say whether it is concave up or down.

Answer _____

2 State m and c for $y = 4 - 2x$. 11.5

Formula _____

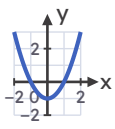
Substitute _____

Answer _____

3 Does $(-2, 7)$ lie on $y = 2x^2 - 1$? Show your working.

Answer _____

4 Equation



Answer _____

5 Solve $2x + 1 = 7$. 11.7

Answer _____

6 Find the x-intercepts of $y = x^2 - 16$.

Formula _____

Substitute _____

Answer _____

7 Which is narrower: $y = 0.5x^2$ or $y = 2x^2$? Why?

Answer _____

8 Find the y-intercept of $y = 3x - 8$. 11.4

Formula _____

Substitute _____

Answer _____

9 The point (2, 1) lies on $y = x^2 + c$. Find c and the vertex.

Formula _____

Substitute _____

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 The arch of a bridge is modelled by $y = -0.5x^2 + 8$ (in metres). How high is the top of the arch, and how wide is the arch at ground level?

Answer _____

2 A satellite dish has the shape of $y = 0.25x^2$ (in cm). How deep is the dish 6 cm out from its centre?

Answer _____

3 Sketch $y = x^2$ and $y = x^2 + 3$ on the same axes, using $x = -2$ to 2. Describe how the second graph is related to the first.

Answer _____

11.10 Parabolas in real life

EASY

Learning intention We are learning to use parabolas to model real situations, such as falling objects and arches.

Success criteria
- ☐ I can complete a table of values for a quadratic model
- ☐ I can read values from the graph of a model
- ☐ I can find the start, the highest point and when the model reaches a value

NOTES — Key ideas for this lesson

TEACHER COPY

KEY TERMS

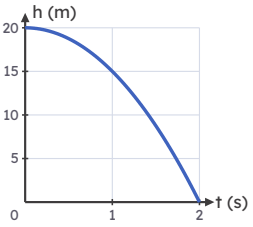
model: an equation that describes a real situation
y-intercept: the starting value (t = 0)
vertex: the highest or lowest point
x-intercepts: where the height is 0 (on the ground)
sensible values: only the times or lengths that make sense, e.g. $t \geq 0$

FORMULAS

Falling stone: $h = -5t^2 + c$ (c = starting height)
Start: put t = 0
On the ground: put h = 0 and solve
Area of a square: $A = x^2$

STEPS

1. substitute to make a table
2. plot and join with a smooth curve
3. read the value asked for
4. check by substituting



WE DO — With your teacher: copy what they write on the board.

Example 1 A stone is dropped from a cliff. Its height after t seconds is $h = -5t^2 + 20$ metres.

a Table

$h = -5t^2 + 20$

t	0	1	2
h			

b What is the h-intercept, and what does it mean?

c Find the height after 0.5 seconds.

d When does the stone hit the ground?

YOU DO — On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER Square t first, then × −5, then + the starting height · on the ground means h = 0

ROUND 1 — Questions 1–4: use each rule to find the value.

1 $h = -5t^2 + 45$, when $t = 2$

Rule $h = -5t^2 + 45$
Substitute
Answer

2 $h = -5t^2 + 80$, when $t = 3$

Rule $h = -5t^2 + 80$
Substitute
Answer

3 $A = x^2$, when $x = 7$

Rule $A = x^2$
Substitute
Answer

4 $d = 5t^2$, when $t = 4$

Rule $d = 5t^2$
Substitute
Answer

ROUND 2 — Questions 5–8: complete each table of values.

5 $h = -5t^2 + 45$

t	0	1	2	3
h				

Square
Working
Values

6 $d = 5t^2$

t	0	1	2	3
d				

Square
Working
Values

7 $A = x^2$

x	1	2	3	4
A				

Square
Working
Values

8 $h = -t^2 + 9$

t	0	1	2	3
h				

Square
Working
Values

11.10 Parabolas in real life

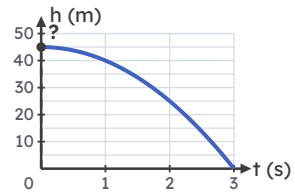
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Read across from the axis to the curve · to find t: put in the value, then solve $t^2 = \ldots$, keep $t \geq 0$.

ROUND 3 Questions 9–11: $h = -5t^2 + 45$. Read the height marked ? from the graph.

9

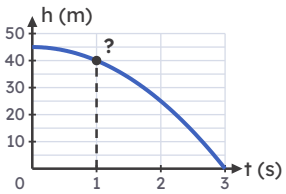


Read _____

Check _____

Answer _____

10

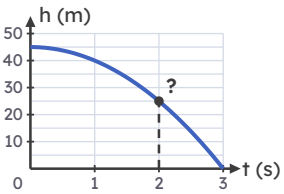


Read _____

Check _____

Answer _____

11



Read _____

Check _____

Answer _____

ROUND 4 Questions 12–14: find when the model reaches the value.

12 $h = -5t^2 + 80$; when is $h = 0$?

Equation _____

Solve _____

Answer _____

13 $h = -5t^2 + 45$; when is $h = 25$?

Equation _____

Solve _____

Answer _____

14 $d = 5t^2$; when is $d = 125$?

Equation _____

Solve _____

Answer _____

ROUND 5 Questions 15–17: use the area model.

15 $A = x^2$; find x when $A = 64$

Equation _____

Solve _____

Answer _____

16 $A = 2x^2$; find A when $x = 5$

Equation _____

Solve _____

Answer _____

17 $A = 2x^2$; find x when $A = 72$

Equation _____

Solve _____

Answer _____

ROUND 6 Questions 18–20: when does each dropped object hit the ground?

18 $h = -5t^2 + 20$

$h = 0$ _____

Solve _____

Answer _____

19 $h = -5t^2 + 180$

$h = 0$ _____

Solve _____

Answer _____

20 $h = -5t^2 + 31.25$

$h = 0$ _____

Solve _____

Answer _____

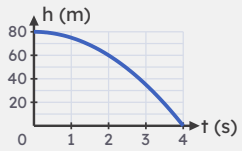
11.10 Parabolas in real life

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 A ball is dropped from 80 m. Its height after t seconds is $h = -5t^2 + 80$ metres.

a Graph



b Find the height after 2.5 seconds.

c When is the ball 35 m above the ground?

d How far does it fall in the first 2 seconds?

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

1 A stone is dropped: $h = -5t^2 + 125$. Find its height after 3 seconds.

Formula _____
Substitute _____
Answer _____

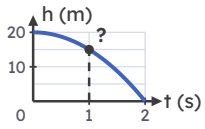
2 Find the vertex of $y = -x^2 + 6$. [11.9]

Answer _____

3 When does the stone in question 1 hit the ground?

Formula _____
Substitute _____
Answer _____

4 Read



Answer _____

5 A plumber charges $C = 60h + 90$ dollars. Find C for 2 hours. [11.8]

Formula _____
Substitute _____
Answer _____

6 A square tile has area $A = x^2$ cm². What side length gives an area of 144 cm²?

Formula _____
Substitute _____
Answer _____

7 Which is higher after 1 second: $h = -5t^2 + 50$ or $h = -4t^2 + 48$?

Answer _____

8 Find the equation of the line through (0, 2) and (4, 10). [11.6]

Formula _____
Substitute _____
Answer _____

9 A dropped stone falls $d = 5t^2$ metres in t seconds. How far does it fall between $t = 1$ and $t = 3$?

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A diver jumps from a 20 m cliff. Her height above the water after t seconds is $h = -5t^2 + 20$ metres. How long until she reaches the water? How high is she after 1.5 seconds?

Answer _____

2 A garden arch is modelled by $y = -0.25x^2 + 4$ (in metres). Find its height and its width at ground level. Can a 3 m tall cart pass under it 1.8 m from the centre?

Answer _____

3 A square photo has area $A = x^2$ cm². A frame needs a photo with an area of 50 cm². Find the side length, correct to one decimal place.

Answer _____